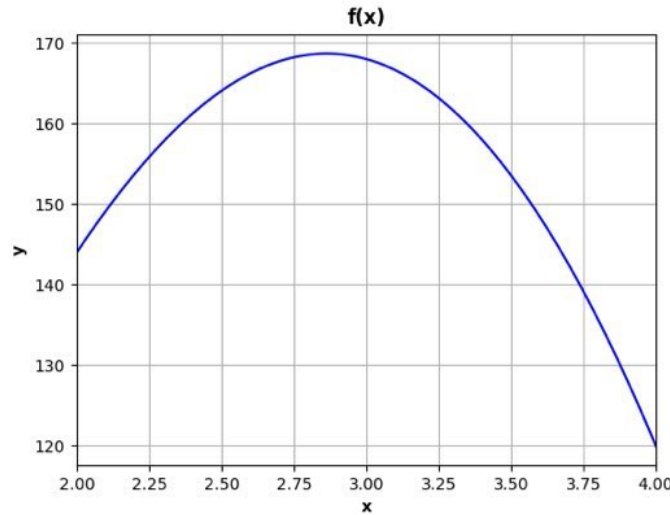


1. **(Numerical Integration)** Approximate the following integral using Simpson's rule and the Trapezoid rule. Use the same three points for the Simpson's rule and Trapezoid rule (using 2 trapezoids). **Show your work.**

$$\int_2^4 f(x) dx$$



Simpson's rule

$$\int_2^4 f(x) dx = \frac{1}{6} \Delta x \left[f(a) + 4f\left(\frac{a+b}{2}\right) + f(b) \right] = \frac{1}{3} (144 + 4 * 168 + 120) = 312.0$$

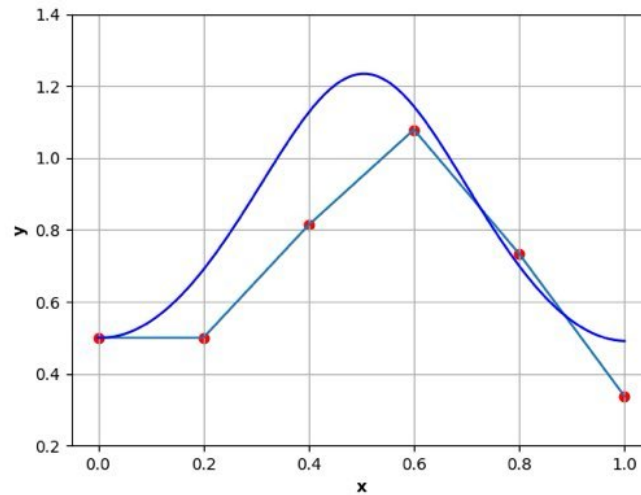
Trapezoid rule

$$\int_2^4 f(x) dx = \frac{1}{2} \Delta x \sum_{i=1}^{n-1} [f(x_i) + f(x_{i+1})] = \frac{1}{2} * (144 + 168) + \frac{1}{2} * (168 + 120) = 300.0$$

2. **(Explicit Integration)** Solve the following differential equation using explicit integration:

$$f(x, y) = \frac{dy}{dx} = \sin(2\pi x) \cdot e^y$$

given the initial condition $(x, y) = (0, 0.5)$ and $\Delta x = 0.2$. **Plot** the resulting function alongside the real result below. You need to plot all the calculated (x_i, y_i) in the figure and draw the slope $\left(\frac{dy}{dx}\right)$ at each (x_i, y_i) .



Use Euler's Method (5 steps)

x	y	$f(x, y)$
0	0.5	0
0.2	0.5	1.568
0.4	0.814	1.326
0.6	1.079	-1.729
0.8	0.733	-1.980
1.0	0.337	0

$$y_{n+1} = y_n + \Delta x \cdot \frac{dy}{dx}$$

$$x_{n+1} = x_n + \Delta x$$

3. **(Derivatives)** Calculate the gradient of the explicit function $\nabla\phi(-2,0)$ represented by the sample points in the grid using 2nd order finite differences. The gradient of a function is defined as its partial derivative along each axis:

$$\nabla\phi = \begin{bmatrix} d\phi/dx \\ d\phi/dy \end{bmatrix}$$

$\nabla\phi = \begin{bmatrix} d\phi/dx \\ d\phi/dy \end{bmatrix}$		-4	-3	-2	-1	0	1	2	3	4
	-3	3.2	2.1	1.8	0.8	1.0	0.8	1.8	2.1	3.2
	-2	2.2	1.8	0.7	0.4	0.0	0.4	0.7	1.8	2.2
	-1	2.1	1.4	0.4	-0.7	-1.0	-0.7	0.4	1.4	2.1

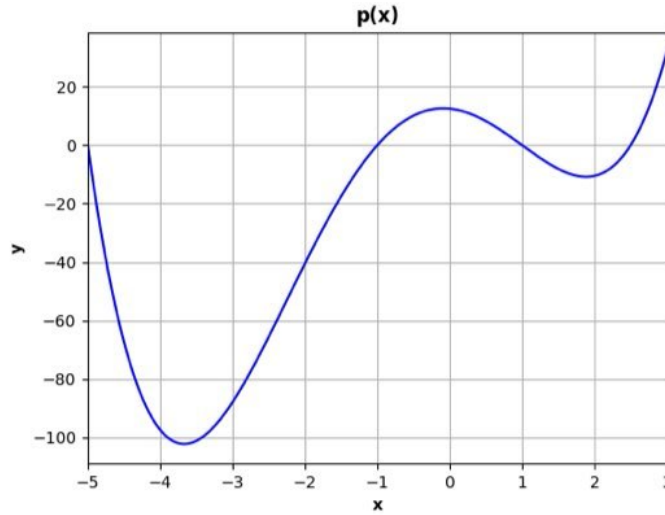
	0	2.0	1.0	0.0	-1.0	-2.0	-1.0	0.0	1.0	2.0
	1	2.1	1.4	0.4	-0.7	-1.0	-0.7	0.4	1.4	2.1
	2	2.2	1.8	0.7	0.4	0.0	0.4	0.7	1.8	2.2

$$f'(x) \approx \frac{f(x + \Delta x) - f(x - \Delta x)}{2\Delta x}$$

$$\frac{d\phi}{dx}(-2,0) = \frac{\phi(-2+1,0) - \phi(-2-1,0)}{2*1} = \frac{\phi(-1,0) - \phi(-3,0)}{2} = \frac{-1.0 - 1.0}{2} = -1.0$$

$$\frac{d\phi}{dy}(-2,0) = \frac{\phi(-2,0+1) - \phi(-2,0-1)}{2*1} = \frac{\phi(2,1) - \phi(2,-1)}{2} = \frac{0.4 - 0.4}{2} = 0.0$$

4. **(Roots of Equations)** Newton's Method. The following is a plot of the polynomial $p(x) = (x + 5)(x - 2.5)(x + 1)(x - 1)$:



- Calculate the root using **3 iterations** of Newton's method given $x_0 = -3.0$ given the expanded polynomial:

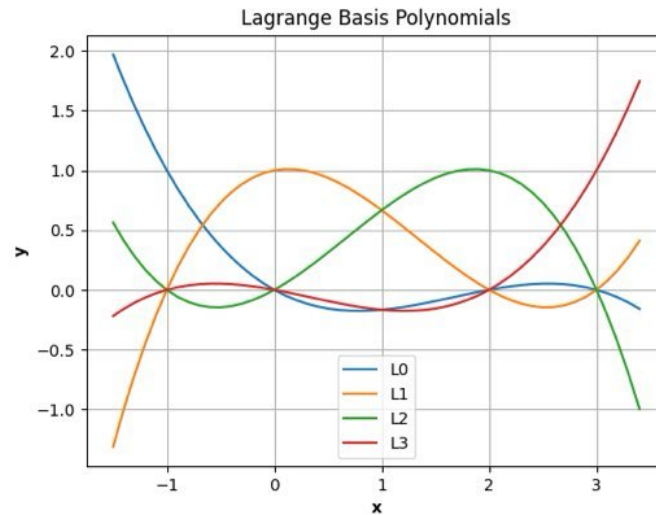
$$x^4 + 2.5x^3 - 13.5x^2 - 2.5x + 12.5$$

Fill the table below and include the **relative** error at each step (the real root is -1). At least Keep 3 significant digits. Illustrate how to determine the x_n on the figure also.

x_n	$f(x_n)$	$f'(x_n)$	x_{n+1}	E_r
-3.0	-88.0	18.20	-0.6842	31.58%
-0.6842	7.310	18.20	-1.086	8.6%
-1.086	-2.510	30.54	-1.003	0.3%
-1.003	-0.099	28.11	-0.9995	0.05%

5. **(Interpolation)** A student is trying to use Lagrange Polynomials to fit a set of nodes (x_i, y_i) , and has created the Lagrange Basis Polynomials in accordance with x . If $y = [-1, 3, 5, 2]$, please

- Show the expressions of all the Lagrange Basis Polynomials of this interpolation in production form.
- Show the Lagrange interpolation of the nodes. Please expand the polynomial.



It is obvious $x = [-1, 0, 2, 3]$, we know

$$L_0(x) = \frac{x}{(-1-0)} \frac{x-2}{(-1-2)} \frac{x-3}{(-1-3)}, L_1(x) = \frac{x+1}{(0+1)} \frac{x-2}{(0-2)} \frac{x-3}{(0-3)}, L_2(x) = \frac{x+1}{(2+1)} \frac{x}{(2-0)} \frac{x-3}{(2-3)}, \text{ and}$$

$$L_3(x) = \frac{x+1}{(3+1)} \frac{x}{(3-0)} \frac{x-2}{(3-2)}. \text{ And,}$$

$$p(x) = \sum y_i L_i(x) = -\frac{1}{12}x^3 - \frac{11}{12}x^2 + \frac{19}{6}x + 3$$

6. **(Least square)** Fit the following points to a cubic polynomial $p(x) = a + bx + cx^2 + dx^3$

x	-1.0	-0.5	0	0.5	1	1.5	2
y	0.1	2.2	3.1	2.7	1.9	1.9	3.1

please

- Show the cost function using least square fitting.
- Find the set of linear equations that can minimize the cost function.
- Solve the linear equations using Gaussian Eliminations to get a, b, c and d . Show your work.
- Predict the y at point $x = -1.5$ and point $x = 0.2$ respectively, using your cubic polynomial.

The cost function using least square fitting is defined as

$$K(\theta) = \sum (p(x_i) - y_i)^2$$

To minimize $K(\theta)$, we have

$$\frac{\partial K(\theta)}{\partial a} = \sum_{i=1}^N 2[a + bx_i + cx_i^2 + dx_i^3 - y_i] = 0$$

$$\frac{\partial K(\theta)}{\partial b} = \sum_{i=1}^N 2[a + bx_i + cx_i^2 + dx_i^3 - y_i] * x_i = 0$$

$$\frac{\partial K(\theta)}{\partial c} = \sum_{i=1}^N 2[a + bx_i + cx_i^2 + dx_i^3 - y_i] * x_i^2 = 0$$

$$\frac{\partial K(\theta)}{\partial d} = \sum_{i=1}^N 2[a + bx_i + cx_i^2 + dx_i^3 - y_i] * x_i^3 = 0$$

Then

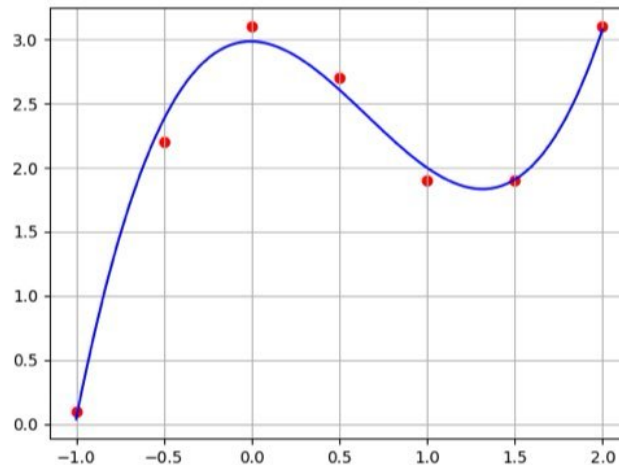
$$\begin{bmatrix} N & \sum x_i & \sum x_i^2 & \sum x_i^3 \\ \sum x_i & \sum x_i^2 & \sum x_i^3 & \sum x_i^4 \\ \sum x_i^2 & \sum x_i^3 & \sum x_i^4 & \sum x_i^5 \\ \sum x_i^3 & \sum x_i^4 & \sum x_i^5 & \sum x_i^6 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} = \begin{bmatrix} \sum y_i \\ \sum y_i x_i \\ \sum y_i x_i^2 \\ \sum y_i x_i^3 \end{bmatrix}$$

Substitute x_i and y_i , we have

$$\begin{bmatrix} 7.0 & 3.5 & 8.75 & 11.38 \\ 3.5 & 8.75 & 11.38 & 23.19 \\ 8.75 & 11.38 & 23.19 & 39.59 \\ 11.38 & 23.19 & 39.59 & 77.42 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} = \begin{bmatrix} 15 \\ 11.1 \\ 19.9 \\ 33.08 \end{bmatrix}$$

Solve the equation using Gaussian Elimination, we got

$$[a \ b \ c \ d] = [2.99 \ -0.02 \ -1.97 \ 1.00]$$



We then know $p(x) = 1.00 * x^3 - 1.97 * x^2 - 0.02 * x + 2.99$, and $p(-1.5) = -4.79$, $p(0.2) = 2.92$.